

Commerce XI Maths App

Exercise 3.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 3 - Skewness | Total questions: 10

Question 1 - ENTER ANSWER · PG-43 · EX-Q1

For a distribution, mean = 100, mode = 127 and SD = 60.
Find the Pearson coefficient of skewness Sk_p .

Correct answer: -0.45

Explanation and steps

$$Sk_p = \frac{100 - 127}{60} = \frac{-27}{60} = -0.45.$$

Question 2 - ENTER ANSWER · PG-43 · EX-Q2

The mean and variance of the distribution is 60 and 100 respectively.
Find the mode of the distribution if $Sk_p = -0.3$.

Correct answer: 63

Explanation and steps

$$S.D. = 10. \text{ Thus } -0.3 = \frac{60 - \text{Mode}}{10}, \text{ so Mode} = 63.$$

Question 3 - ENTER ANSWER · PG-43 · EX-Q2

The mean and variance of the distribution is 60 and 100 respectively.
Find the median of the distribution if $Sk_p = -0.3$.

Correct answer: 61

Explanation and steps

Mean - Mode = $3 \times (\text{Mean} - \text{Median})$. With mode 63: $-3 = 3 \times (60 - \text{Median})$, hence Median = 61.

Question 4 - ENTER ANSWER · PG-43 · EX-Q3

For a data set, sum of upper and lower quartiles is 100,
difference between upper and lower quartiles is 40 and median is 50.
Find the coefficient of skewness.

Correct answer: 0

Explanation and steps

$$Sk_b = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} = \frac{100 - 100}{40} = 0. \text{ The printed textbook answer 1 is inconsistent with its own data and formula.}$$

Question 5 - ENTER ANSWER · PG-43 · EX-Q4

For a data set with upper quartile equal to 55 and median equal to 42. If the distribution is symmetric, find the value of lower quartile.

Correct answer: 29

Explanation and steps

For symmetry, $Q_1 + Q_3 = 2Q_2$. Thus $Q_1 = 84 - 55 = 29$.

Question 6 - ENTER ANSWER · PG-43 · EX-Q5

Height (inches)	Less than 60	60–64	64–68	68–72	72–76
No. of females	10	20	40	10	2

Obtain coefficient of skewness by formula.

Correct answer: -0.1881

Explanation and steps

Using the grouped distribution gives a Pearsonian coefficient of approximately -0.1881 .

Question 7 - MCQ · PG-43 · EX-Q5

Height (inches)	Less than 60	60–64	64–68	68–72	72–76
No. of females	10	20	40	10	2

Comment on nature of the distribution.

Options

- A. negatively skewed
- B. positively skewed
- C. symmetric
- D. cannot be determined

Correct answer: negatively skewed

Explanation and steps

The coefficient is negative, so the distribution is negatively skewed.

Question 8 - ENTER ANSWER · PG-43 · EX-Q6

Find Sk_p for the following set of observations.

17, 17, 21, 14, 15, 20, 19, 16, 13, 17, 18

Correct answer: 0

Explanation and steps

Mean = Median = Mode = 17, hence $Sk_p = 0$.

Question 9 - ENTER ANSWER · PG-43 · EX-Q7

Calculate Sk_b for the following set of observations of yield of wheat in kg from 13 plots:

4.6, 3.5, 4.8, 5.1, 4.7, 5.5, 4.7, 3.6, 3.5, 4.2, 3.5, 3.6, 5.2

Correct answer: -0.5

Explanation and steps

After ordering, $Q_1 = 3.6$, $Q_2 = 4.6$ and $Q_3 = 4.8$. Thus $Sk_b = \frac{4.8 + 3.6 - 9.2}{4.8 - 3.6} = -0.5$.

Question 10 - ENTER ANSWER · PG-43 · EX-Q8

For a frequency distribution, $Q_3 - Q_2 = 90$ and $Q_2 - Q_1 = 120$.
Find Sk_b .

Correct answer: -0.143

Explanation and steps

$$Sk_b = \frac{90 - 120}{90 + 120} = \frac{-30}{210} = -0.142857 \approx -0.143.$$